

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 2: CELL: THE BUILDING BLOCK OF LIFE [SET C - NCERT EXEMPLAR SPECIAL]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH02/SET-C****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS (SET C - NCERT EXEMPLAR & CRITICAL THINKING):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** NCERT Exemplar based Multiple Choice Questions with single correct option.
3. **Section B (Q16 – Q20):** Deep Assertion-Reasoning based questions testing scientific cause-and-effect.
4. **Section C (Q21 – Q25):** Case-Study / Laboratory Experimental Investigation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: NCERT EXEMPLAR MCQS (Q.1 TO Q.15)**[15 MARKS]****Q.1 Why are bacterial cells killed when placed in highly concentrated salt brine (pickles) or concentrated sugar syrup (jams)?**

- (A) Salt chemically digests the bacterial capsule (B) Severe exosmosis causes plasmolysis, causing cellular dehydration and death of bacteria
- (C) Sugar blocks bacterial flagella (D) Bacteria absorb too much water and burst

Q.2 Which of the following is considered strong evolutionary evidence supporting the endosymbiotic origin of mitochondria and chloroplasts?

- (A) They are surrounded by a single membrane identical to the ER (B) They contain circular double-stranded DNA and 70S ribosomes similar to prokaryotes
- (C) They synthesize glycogen like animal cells (D) They lack any proteins

Q.3 A plant cell has a turgor pressure equal in magnitude to its osmotic pressure. The net water movement across the plasma membrane is:

- (A) Rapid continuous influx (B) Zero net movement (dynamic equilibrium)
- (C) Rapid continuous efflux (D) Immediate cell wall dissolution

Q.4 Lysosomal hydrolytic enzymes are synthesized on ribosomes attached to the:

- (A) Smooth Endoplasmic Reticulum (B) Rough Endoplasmic Reticulum
- (C) Free cytoplasmic polysomes only (D) Mitochondrial matrix

Q.5 Which of the following cellular structures lacks a surrounding lipid membrane entirely?

- (A) Lysosome (B) Centrosome / Centriole
- (C) Peroxisome (D) Vacuole

Q.6 The chromosome region where the two sister chromatids are held together and where the spindle fibers attach during cell division is the:

- (A) Centrosome (B) Centromere (Kinetochore)
- (C) Chromomere (D) Telomere

Q.7 What is the primary role of the contractile vacuole in freshwater unicellular organisms like *Amoeba*?

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|--------------------------------|--|
| (A) Digestion of engulfed food | (B) Osmoregulation (pumping out excess incoming water) |
| (C) Cellular respiration | (D) Lipid storage |

Q.8 Which organelle is often referred to as the 'traffic controller' or 'director of macromolecular dispatch' in eukaryotic cells?

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|---------------------|---------------|
| (A) Golgi apparatus | (B) Ribosome |
| (C) Chloroplast | (D) Centriole |

Q.9 A cell undergoing mitosis is treated with a chemical colchicine that destroys the microtubule spindle fibers. At which stage will mitosis be arrested?

- | | |
|----------------|-----------------|
| (A) Interphase | (B) Metaphase |
| (C) Telophase | (D) Cytokinesis |

Q.10 In plant cells, cytokinesis (division of cytoplasm) takes place by:

- | | |
|--|--|
| (A) Cleavage furrow forming from periphery to center | (B) Cell plate formation proceeding centrifugally from center outward to periphery |
| (C) Simple budding | (D) Disintegration of the cell wall |

Q.11 Which organelle contains enzymes that detoxify hydrogen peroxide (H_2O_2) into water and oxygen using catalase?

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|----------------|---------------|
| (A) Peroxisome | (B) Lysosome |
| (C) Ribosome | (D) Nucleolus |

Q.12 Chromoplasts contain which fat-soluble pigments that impart bright orange, red, and yellow colors to flowers and fruits?

- | | |
|-------------------------|--|
| (A) Chlorophyll a and b | (B) Carotenoids (carotenes and xanthophylls) |
| (C) Anthocyanins only | (D) Melanin |

Q.13 How does prokaryotic DNA differ fundamentally from eukaryotic nuclear DNA?

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|--|---|
| (A) Prokaryotic DNA is single-stranded RNA | (B) Prokaryotic DNA is circular, naked (lacks histone proteins), and not enclosed in a nuclear membrane |
| (C) Prokaryotic DNA contains no adenine | (D) Prokaryotic DNA is located inside vacuoles |

Q.14 The fluid-mosaic model of the plasma membrane was proposed by:

- | | |
|----------------------------------|--------------------------------|
| (A) Robert Hooke and Leeuwenhoek | (B) Singer and Nicolson (1972) |
| (C) Watson and Crick | (D) Schleiden and Schwann |

Q.15 Which of the following processes requires cellular energy expenditure in the form of ATP?

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|--|--|
| (A) Diffusion of carbon dioxide | (B) Osmosis of water through aquaporins |
| (C) Active transport of sodium and potassium ions (Na^+/K^+ pump) | (D) Facilitated diffusion of glucose down concentration gradient |

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): The nuclear envelope is interrupted by numerous nuclear pores.

Reason (R): Nuclear pores allow bidirectional transport of RNA and protein molecules between the nucleoplasm and cytoplasm.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): Red blood cells burst when placed in distilled water, but onion epidermal cells do not.

Reason (R): Onion epidermal cells have a tough outer cellulose cell wall that resists osmotic burst pressure.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): Gametes produced by meiosis are genetically diverse.

Reason (R): Crossing over between homologous chromosomes and independent assortment of maternal/paternal chromosomes occur during meiosis.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): The inner membrane of mitochondria is more permeable than its outer membrane.

Reason (R): The inner mitochondrial membrane contains porin proteins that form large open channels.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) Both (A) and (R) are false.

Q.20 Assertion (A): Endocytosis does not occur in plant cells.

Reason (R): Plant cells are enveloped by a rigid, non-flexible cellulose cell wall that prevents membrane invagination.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

SECTION C: EXPERIMENTAL INVESTIGATION & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL INVESTIGATION:

Case Study: Artificial Cell Osmosis Model using Dialysis Tubing

In an experiment investigating membrane permeability, students fill a cellulose dialysis tubing bag with a mixture of 10% starch solution and 5% glucose solution. The bag is tied securely at both ends and submerged into a beaker containing distilled water to which yellow iodine solution has been added. After 45 minutes, students observe that:

1. The liquid inside the dialysis bag turns deep blue-black.
2. The solution outside in the beaker remains light yellow.
3. Testing the beaker liquid with Benedict's reagent yields an orange-red precipitate upon boiling.

Q.21 Why did the liquid inside the dialysis bag turn blue-black?

- (A) Starch moved out of the bag into the beaker
 (B) Iodine molecules diffused across the pores into the bag, reacting with starch
 (C) Glucose decomposed into iodine
 (D) The dialysis membrane produced dye

Q.22 Why did the solution in the beaker outside NOT turn blue-black?

- (A) Starch molecules are too large to pass through the microscopic pores of the dialysis membrane
 (B) Iodine repels starch chemically
 (C) Starch decomposed into pure carbon
 (D) Water boiled away

Q.23 What does the positive Benedict's test in the beaker solution prove?

- (A) Dialysis tubing is permeable to small glucose molecules
 (B) Starch was converted into protein
 (C) Iodine contains reducing sugars
 (D) The beaker is made of sugar

Q.24 In this experiment, what physical property governed whether a molecule crossed the membrane?

- (A) Molecular size / diameter relative to pore dimensions
 (B) Color of the molecule
 (C) Whether the molecule was organic or inorganic
 (D) Atmospheric pressure

Q.25 What change would you expect in the volume and firmness of the dialysis bag after 45 minutes?

- (A) The bag shrivels and collapses
 (B) The bag swells and becomes turgid due to osmotic influx of water
 (C) The bag dissolves completely
 (D) Volume remains unchanged

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET C)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & EXEMPLAR EXPLANATIONS (SET C)

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.1	(B)	Hypertonic brines and syrups draw water out of bacterial and fungal cells via exosmosis (plasmolysis), arresting metabolism and killing contaminants.
Q.2	(B)	Like free-living prokaryotes, mitochondria and chloroplasts contain naked circular DNA, 70S ribosomes, and divide independently via binary fission.
Q.3	(B)	When turgor pressure equals osmotic pressure ($\text{STP} = \text{OP}$), the diffusion pressure deficit ($\text{DPD} = \text{OP} - \text{TP} = 0$) becomes zero; net osmotic flow ceases.
Q.4	(B)	Hydrolases are synthesized by membrane-bound ribosomes on the Rough ER, then transported to the Golgi apparatus for sorting into primary lysosomes.
Q.5	(B)	Centrosomes (and centrioles) are non-membrane-bound barrel-shaped microtubule organizing centers found in animal cells. Ribosomes and nucleoli also lack membranes.
Q.6	(B)	The centromere is the primary constriction holding sister chromatids together, containing kinetochore protein complexes where mitotic spindle microtubules anchor.
Q.7	(B)	Freshwater is hypotonic to Amoeba cytoplasm. The contractile vacuole continuously collects excess endosmotic water and expels it to prevent lysis.
Q.8	(A)	The Golgi apparatus chemically modifies, packages, sorts, and directs secretory and membrane proteins to their appropriate cellular destinations.
Q.9	(B)	Colchicine disrupts spindle fiber polymerization, preventing sister chromatids from separating; cells are arrested at metaphase.
Q.10	(B)	In rigid plant cells, Golgi vesicles fuse in the center to form a cell plate (phragmoplast), expanding centrifugally outward to meet the existing wall.
Q.11	(A)	Peroxisomes contain catalase and oxidases that break down toxic hydrogen peroxide ($2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$) generated during cellular oxidation.
Q.12	(B)	Chromoplasts synthesize and store lipid-soluble carotenoid pigments (yellow xanthophylls and orange-red carotenes) to attract pollinators and seed dispersers.
Q.13	(B)	Prokaryotes carry a single circular double-stranded DNA chromosome devoid of histone protein packaging, situated in an unmembrane-bound nucleoid.
Q.14	(B)	S.J. Singer and G.L. Nicolson proposed the fluid-mosaic model, describing membrane proteins floating in or on a fluid phospholipid bilayer.
Q.15	(C)	Active transport moves ions against their electrochemical concentration gradient, requiring metabolic ATP hydrolysis.
Q.16	(A)	Nuclear pore complexes regulate the entry of nuclear proteins (DNA polymerases, histones) and the exit of mRNA transcripts and ribosomal subunits.
Q.17	(A)	The rigid plant cell wall exerts wall pressure equal to turgor pressure, halting net influx and preventing lysis, whereas animal RBCs lack a cell wall.
Q.18	(A)	Recombination during prophase I and random chromosome alignment in metaphase I generate novel gene combinations, ensuring genetic variation.
Q.19	(D)	Both statements are false: The *outer* membrane contains porins and is highly permeable; the *inner* membrane is strictly selectively impermeable to maintain the proton electrochemical gradient.
Q.20	(A)	The rigid cell wall prevents the plasma membrane from performing large-scale invagination or engulfment of particulate matter as seen in amoeboid animal cells.
Q.21	(B)	Iodine molecules are small and readily diffuse through the dialysis membrane pores into the bag, forming a dark blue-black complex with starch.

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.22	(A)	Starch is a high-molecular-weight polymer that cannot cross the semi-permeable dialysis membrane pores.
Q.23	(A)	Small monosaccharide glucose molecules diffuse out of the dialysis bag through the membrane pores into the surrounding water.
Q.24	(A)	Dialysis tubing functions as a size-exclusion semi-permeable membrane: small molecules (iodine, glucose, water) pass; large macromolecules (starch) are retained.
Q.25	(B)	Because the bag interior has a high solute concentration (hypertonic) relative to distilled water outside, water enters by endosmosis, increasing turgor.